

GDR QCD Nantes 2026

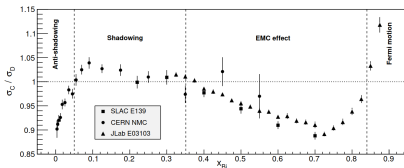
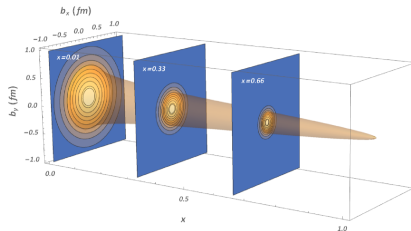
Coherent DVCS off ^4He with
ALERT and CLAS at JLab

Felix Touchte Codjo

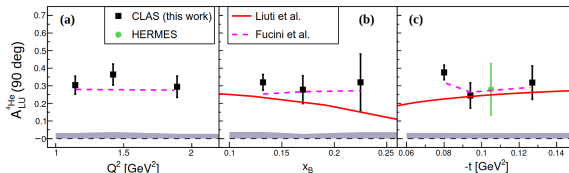
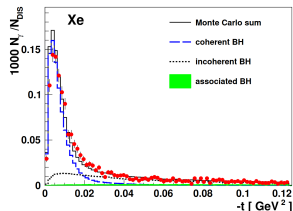
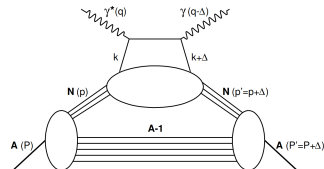
IJCLab

October 7, 2026

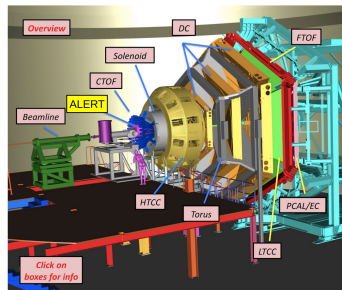
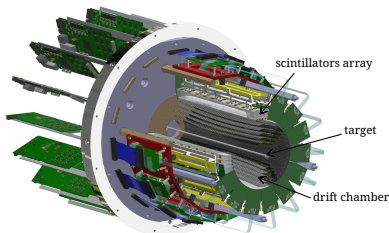
- ▶ The Generalized Parton Distributions (GPDs) combine the notion of PDFs and FFs, along with their correlations
- ▶ We can access GPDs off the free proton or off a nucleus
- ▶ A direct comparison of these distributions can reveal a variety of nuclear effects
 - e.g. modification of the quark confinement size, short range correlation
- ▶ This will provide new insights into the understanding of the EMC effect



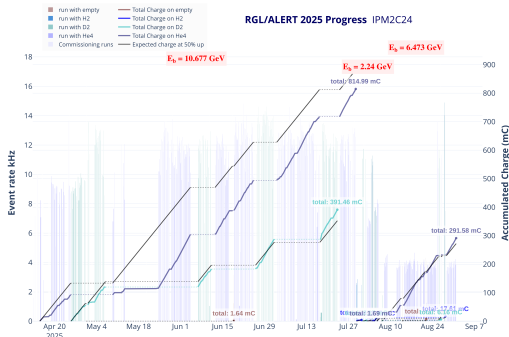
- ▶ We can access quark GPDs in DVCS
- ▶ For a nuclear target, the DVCS can be coherent or incoherent
- ▶ First measurements by HERMES
 - The recoil nuclei were not detected
 - $-t$ based threshold to separate the channels
 - Measurements affected by large uncertainties
 - No nuclear dependence of the BSA found
- ▶ CLAS eg6 : first coherent DVCS off ^4He
 - The recoil nucleus was detected
 - Coherent and incoherent channels were isolated
 - The coherent BSA is twice the one measured on the free proton



- ▶ ALERT is an extension of the CLAS eg6 at higher luminosity and 12 GeV
- ▶ It is based on a new detector, A Low Energy Recoil Tracker
 - Along with the ^4He , identify the two isobars ^3He et ^3H
 - With kinematics: $70 < p < 400 \text{ MeV}/c$, $\theta > 45^\circ$
 - To be placed inside a 5T solenoid magnet
 - To be transparent to the particles detected by CLAS
- ▶ ALERT is a combination of a drift chamber (AHDC) and an array of scintillators (ATOF)



- ▶ The run period took place from April 2025 to September 2025 at JLab
- ▶ Run summary
 - Targets: helium, deuterium, hydrogen
 - Beam energies: 2.2 GeV, 6.4 GeV, 10.6 GeV
 - Beam current: up to 325 nA
 - Event rate: 10 kHz
 - 6.42×10^{10} trigger events collected
- ▶ My activities at JLab
 - Data monitoring
 - On-call detector expert
 - Target change
 - Target purge
- ▶ The data taking was a success

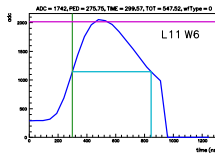
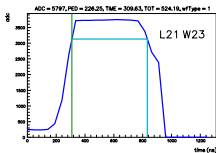
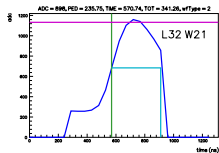
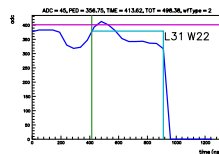
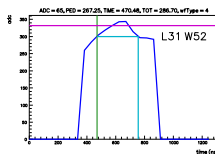
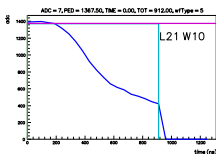
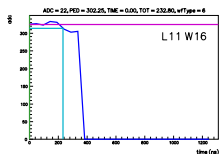


After the end of the run period, efforts mainly focus on the finalization of the reconstruction software and the calibration of the data

- ▶ Decoding
 - From raw data to waveform
 - Waveform processing**
- ▶ Hit selection
 - Waveform classification**
 - Selection cuts**
- ▶ Track finder
 - MLP/GNN
- ▶ Track fitting
 - Helix fitter
 - Kalman Filter*
- ▶ Hit matching : AHDC $\rightleftharpoons$ ATOF
- ▶ Particle identification
- ▶ Calibration
 - Time
 - Time-to-distance*
- ▶ Alignment
 - Vertex alignment**
 - Phi alignment**
 - Beamline orientation
- ▶ Simulation
 - Digitization*
 - Background merging

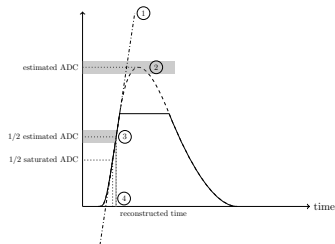
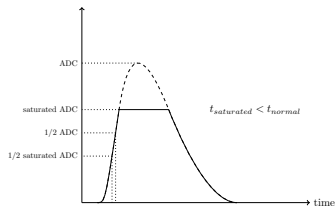
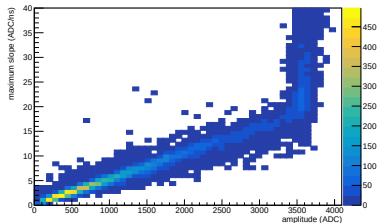
- ▶ The signals recorded by the AHDC have various shapes
- ▶ These different waveforms need specific treatments
- ▶ We gather them into seven categories

wfType 6 → too short wfType 3 → pileup (ongoing work) wfType 0 → normal
 wfType 5 → decreasing wfType 2 → missing trailing edge
 wfType 4 → small ToT wfType 1 → saturated



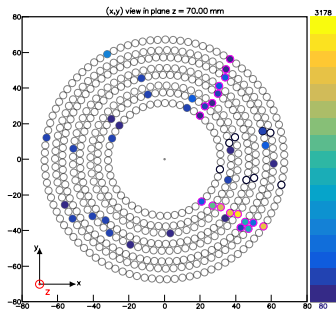
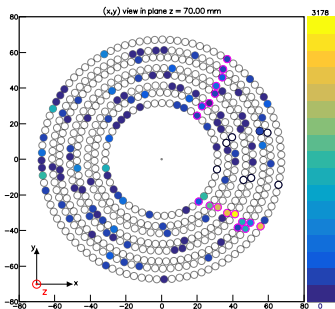
- ▶ Only signals with a good leading edge time are relevant ($wfType \leq 2$)

- ▶ The signals of α particles always saturate
- ▶ The time extracted at 1/2 saturated ADC is under-estimated
 - It badly impacts the reconstruction
- ▶ To correct for the amplitude, we use the linear correlation of the maximum slope of a non-saturated signal with its amplitude
- ▶ With the corrected amplitude, we can recompute the time at 1/2 amplitude



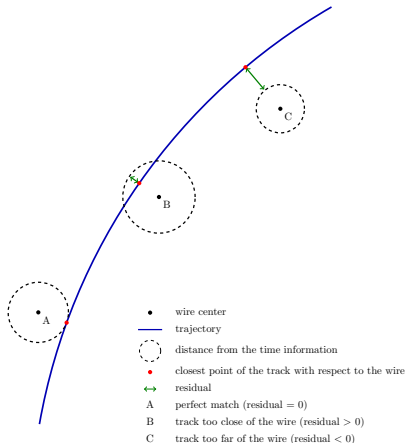
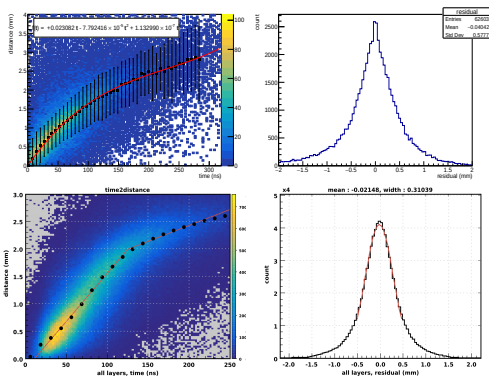
- ▶ This method has been compared with the one based on the time over threshold

- ▶ Selection criteria: $wfType \leq 2$ and time cuts
- ▶ The occupancy in the AHDC is much lower after the hit selection
- ▶ It simplifies the task of the track finder

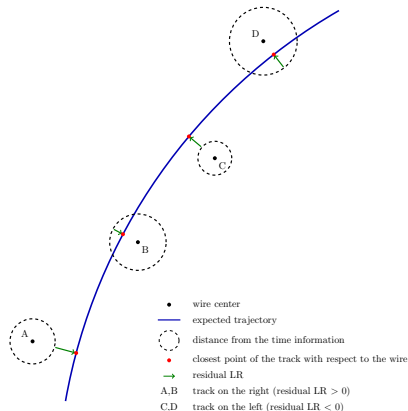
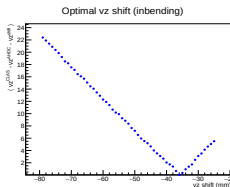
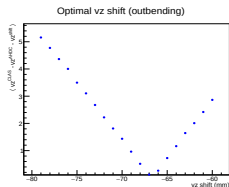
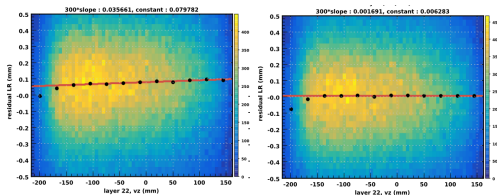


Event display: 325 nA, 10.6 GeV, ^4He target

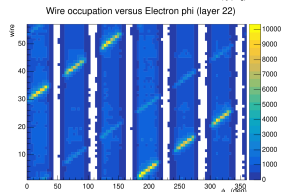
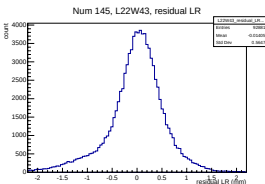
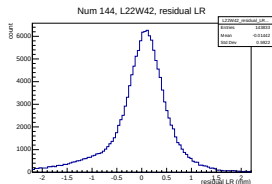
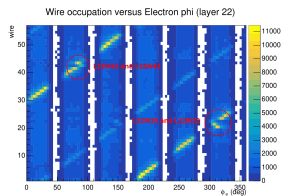
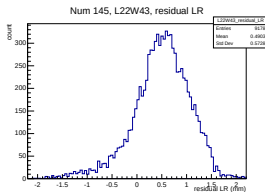
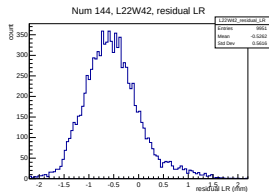
- ▶ The AHDC is a Time Projection Chamber
- ▶ The position is obtained by means of the time-to-distance which is subject to calibration over the run period
- ▶ We use an iterative procedure based on the minimization of the residuals



- ▶ Extract from data the relative disposition of the detector components
- ▶ We use elastic events for which the reconstructed track and the expected track can be compared
 - Phi alignment (by layer or by wire) $\rightarrow$ residual_LR
 - Z alignment with respect to CLAS $\rightarrow$ position scan



- ▶ We found some issues in the translation table of the AHDC
- ▶ Some wire residual distributions were non-centered or had unexpected shapes
- ▶ We found that they were the result of wire swaps as illustrated in the wire occupancy versus the electron phi in elastic events
- ▶ In total, four pairs of wires were reversed



- ▶ All the work on the software and calibration is to prepare for my analysis
- ▶ Warnings
 - The AHDC reconstruction is not completed
 - The calibration is still ongoing (ALERT and CLAS)
 - We made basic selection cuts
 - The BSA has been integrated over Q^2 , x_B and $-t$
 - The errors are only statistical (no π^0 background subtraction)
 - The study has been done on 7.53 % of the dataset
- ▶ However, we can already see an asymmetry

